



Introduction to Sports Analytics

What is Sports Analytics?

Why the current emphasis on Sports Analytics?

Applications of Sports Analytics

What makes Python suitable for Sports Analytics?

About this course

- Module 1 - Introduction to Python
- Module 2 - Cricket Analytics case study
 - Using plotly
 - Image processing
 - Video processing

What is Sports Analytics?

What is Sports Analytics?

As the name suggests it is the analysis of data related to different sports. It includes both on-field and off-field analytics.

On-field analytics focuses on improving the performance of player/team on the field. All data that is relevant to improving coaching strategies, player fitness and the like is analyzed using advanced Machine Learning techniques.

Off-field analytics, often focuses on analyzing data to improve the profitability of the franchise and includes analyzing revenue from stadium, revisiting marketing strategies, fan engagement, acquisition strategies, etc.

Why Sports Analytics Now?

Motivation

- Sports analytics market investments are projected to grow upto \$8.4 billion by 2026. (<https://www.marketsandmarkets.com/Market-Reports/sports-analytics-market-35276513.html>)
- According to the U.S Bureau of Labor Statistics, the median salary of a Sports Statistician in 2018 was \$69,700, and the top 10% of earners made more than \$132,320.
(https://www.purdue.edu/science/careers/what_can_i_do_with_a_major/Career%20Pages/sports_statistician.html)
- Historic game data is analyzed to uncover strategies for player development, optimize play calls in games, enhance fan experience, ticket pricing strategies and so on.



Career Path of a Sports Data Analyst

- As a sports data analyst, you may start as an assistant, then gain more responsibilities and work on more complex projects as an intermediate analyst.
- As an advanced analyst, you may lead teams, develop new data sources and methods, and provide strategic guidance.
- In a managerial/director role, you may oversee a team of analysts, set goals and objectives, and ensure alignment with the organization's goals. You may also develop partnerships and secure funding for new projects.

Sports Analytics Jobs in Market

- **Sports Statistician**
- **Sports Scientist / Data Analyst**
- **Senior / Chief Data Scientist**
- **Team Data Solutions Strategist**
- **Vice President(Analytics)**
- **Director Of Analytics**

Role of a Sports Data Analyst

- ❑ Requires collection and monitoring of performance related statistics with respect to a particular player, team or game.
- ❑ From a business perspective, the analysis should improve the overall performance of the team.
- ❑ From an entertainment perspective, the analytical findings can improve the fan engagement by providing stats in between a game

Required Skills

- ❑ In-depth understanding of sports
- ❑ Relationship management skills
- ❑ Presentation skills
- ❑ Technical skills
- ❑ Data Identification

Why the current emphasis on Sports Analytics?

Technological advancements have made data collection easy today. For eg: IOT devices help us collect ball by ball data for each ODI or T20 game.

Improvements in data storage facilities have made data storage very affordable. Datasets that contain data about all professional cricket players from across the globe can be easily stored on the Cloud.

Advances in web technologies have made this data easily transmissible.

Why the current emphasis on Sports Analytics?

Developments in the field of data sciences and machine technologies enable us to analyze vast amounts of data in a short period of time.

Sports teams are now able to analyze historic player and game data to evaluate their team performance, work on coaching plans, assess ways to increase fan engagement, develop bidding strategies and so on.

Applications of Sports Analytics

Moneyball - How Oakland Athletics went on to win 20 games in a row using data analytics?!

Moneyball And The Rise Of Sports Analytics

The influence of sports analytics has never been greater, but does analytics take the fun out of it?

Steve McCaskill, July 8, 2013, 10:30 am



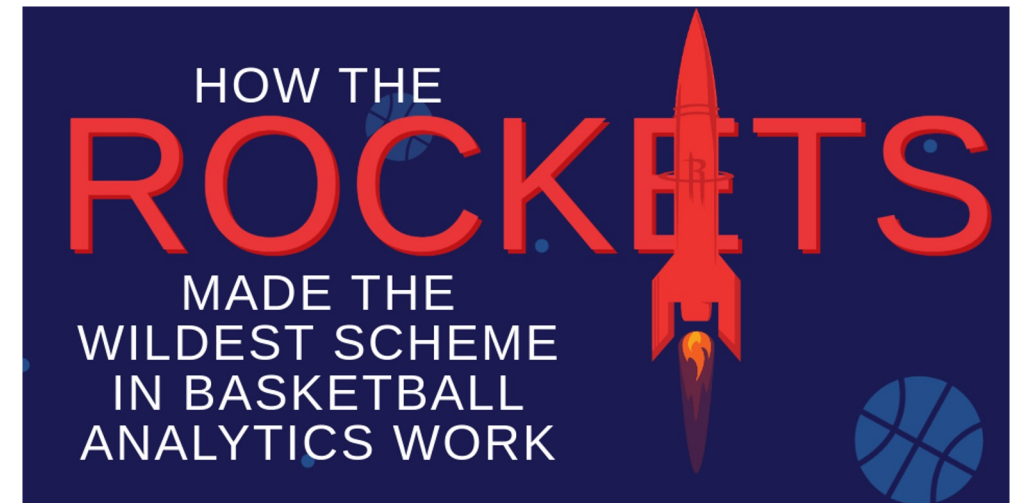
New model could have 'Moneyball'-like impact on baseball players' value



A new machine learning model developed by researchers at the Penn State College of Information Sciences and Technology would completely change, and could enhance, the way the state of a baseball game and a player's impact on the game is measured. The model improves on the existing family of baseball statistics, known as sabermetrics, which measures how many times a player or team achieves a discrete event—such as hitting a double or a home run. The model developed by Penn State researchers views the game as a sequence of events, examining the context that surrounds each individual play—such as whether there were runners on base or if the batter was facing a full count—to get a more nuanced and computational description of how players impact the game. Credit: Adobe Stock: VIAR PRO studio. All Rights Reserved.

Data Science and the Houston Rockets- 3-point shooters

- Daryl Morey, GM of the Houston Rockets team embraced cutting edge analytics in early 2015.
- The analytics team suggested that the team increase the number of their three point shots.
- This led to Houston Rockets having 2017/18 as one of their best seasons, with the highest number of threes along with a higher win/loss ratio



Cricket Analytics: A Game-Changing Revolution

- Cricket analytics is a specialized branch of sports analytics that revolves around the sport of Cricket
- Cricket analytics has revolutionized the way the game is understood, played, and strategized. It has transformed traditional approaches to coaching, decision-making, and player evaluation by introducing a more data-centric and evidence-based methodology
- Cricket analytics has used advanced technology and data analysis to reveal hidden patterns and insights, enhancing the game and transforming the fan experience

Applications of Cricket Analytics

- Player Performance Analysis:
 - Evaluate player strengths and weaknesses
 - Analyze key metrics: batting average, strike rate, bowling economy, and fielding efficiency
 - Provide valuable insights for player improvement and personalized coaching
- Opponent Analysis:
 - Strategize against rival teams and players
 - Analyze historical data and playing styles of opponents
 - Identify weaknesses to adapt game plans and exploit vulnerabilities

Applications of Cricket Analytics

- Talent Identification and Recruitment:
 - Spot and nurture young talent
 - Use data-driven analysis for player recruitment and auctions
 - Invest in the development of promising players
- Injury Prevention and Workload Management:
 - Monitor player fitness and health
 - Manage player workloads to prevent injuries
 - Schedule rest periods during the cricket calendar

Applications of Cricket Analytics

- Match Strategy and Decision Making:
 - Real-time data and insights during matches
 - Tactical planning on field placements, bowling changes, and batting orders
 - Making data-driven decisions to adapt to evolving match situations

ESPN SuperStats

- IIT Madras in association with ESPNcricinfo and Gyan Data developed a data science driven tool called ‘SuperStats’
- The tool is a suite of metrics that helps fans judge performances in limited-overs cricket in a far more nuanced manner than conventional metrics do
- Ramesh Kumar, Vice President and Head, ESPN (India and South Asia) said “The use of data science driven tools in sports have helped us to offer a much more enriching content experience to our discerning fans and cricket enthusiasts”



StanceBeam - Data-driven insights to improve batting!



This Smart Cricket Bat Sensor Technology Can Study Shots Using A Mobile App - Republic World

Posted on 15 November 2019 at 03:26 am

The Sensor called StanceBeam Striker is essentially an Internet of Things (IoT) enabled device. It provides users with useable, real-time and instant 360-degree batting performance data analytics. This data is expected to help players and their coaches study and improve batting performance.

Performance Analysis of Player

- Video motion analysis is a technique used to get information about moving objects from video.
- Video analysis technology paired with smart phones has become more affordable in recent times.
- Each player's performance is analysed to draw a strategy to better their contribution within the team.



About this course

About this course

Module 1 - Introduction to Python

- Introduction to Data Structures
- Control Structures and Looping Statements
- Functions
- NumPy and Pandas libraries for data analysis
- Matplotlib and Seaborn for data visualization

Module 2 - Cricket Analytics case study using plotly visualization.

Module 3 - Cricket Analytics - Image processing

Module 4 - Cricket Analytics - Video processing

```
        operation == "MIRROR_X":
            mirror_mod.use_x = True
            mirror_mod.use_y = False
            mirror_mod.use_z = False
        operation == "MIRROR_Y":
            mirror_mod.use_x = False
            mirror_mod.use_y = True
            mirror_mod.use_z = False
        operation == "MIRROR_Z":
            mirror_mod.use_x = False
            mirror_mod.use_y = False
            mirror_mod.use_z = True

    #selection at the end -add
    mirror_ob.select= 1
    mirror_ob.select=1
    context.scene.objects.active
    ("Selected" + str(modifier_
    mirror_ob.select = 0
    = bpy.context.selected_obj
    data.objects[one.name].select
    print("please select exactly
```

THANK YOU